Commit 5e7f73a



julien-c and LysandreJik committed on Jul 25, 2019

```
[GPT-2] Top-K implementation
cc @LysandreJik

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```

master

1 parent 2c0c615 commit 5e7f73a

3 files changed +114 -7 lines changed

Search within code

CoreMLBert.xcodeproj/project.pbxproj +4 -6

```
@@ -8,6 +8,7 @@

/* Begin PBXBuildFile section */
                791D169322EA4918004D7A79 /* MultiArrayUtilsTests.swift in Sources */ = {isa = PBXBuildFile; fileRef = 791D169222EA4918004D7A79 /* MultiArrayUtilsTests.swift */; };
+               7934AF8822EA4CEC00396AD5 /* gpt2-512.mlmodel in Sources */ = {isa = PBXBuildFile; fileRef = 7934AF8722EA4CEC00396AD5 /* gpt2-512.mlmodel */; };
                7944CDBF22C501D6000C0B1E /* dev-v1.1.json in Resources */ = {isa = PBXBuildFile; fileRef = 7944CDBE22C501D6000C0B1E /* dev-v1.1.json */; };
                7944CDC022C501D6000C0B1E /* dev-v1.1.json in Resources */ = {isa = PBXBuildFile; fileRef = 7944CDBE22C501D6000C0B1E /* dev-v1.1.json */; };
                796DF51022E0EB1D00140C02 /* GPT2Tokenizer.swift in Sources */ = {isa = PBXBuildFile; fileRef = 796DF50F22E0EB1D00140C02 /*
```

```
GPT2Tokenizer.swift */; };
```

```
@@ -61,8 +62,6 @@
```

```
61  62                  79F2CCA022C666C7009F8551 /* question_tokens.json in Resources */ = {isa = PBXBuildFile; fileRef = 79F2CC9F22C666C7009F8551 /* question_tokens.json */; };
62  63                  79F2CCA222C6717E009F8551 /* LoaderView.swift in Sources */ = {isa = PBXBuildFile; fileRef = 79F2CCA122C6717D009F8551 /* LoaderView.swift */; };
63  64                  79F7060E22EA0CA900C4432C /* BERTSQUADFP16.mlmodel in Sources */ = {isa = PBXBuildFile; fileRef = 79F2CC9022C5590C009F8551 /* BERTSQUADFP16.mlmodel */; };
64      -               79F7061022EA472D00C4432C /* gpt2-256.mlmodel in Sources */ = {isa = PBXBuildFile; fileRef = 79F7060F22EA472D00C4432C /* gpt2-256.mlmodel */; };
65      -               79F7061122EA472D00C4432C /* gpt2-256.mlmodel in Sources */ = {isa = PBXBuildFile; fileRef = 79F7060F22EA472D00C4432C /* gpt2-256.mlmodel */; };
66  65              /* End PBXBuildFile section */
67  66
68  67              /* Begin PBXContainerItemProxy section */
```

```
@@ -84,6 +83,7 @@
```

```
84  83
85  84              /* Begin PBXFileReference section */
86  85                  791D169222EA4918004D7A79 /* MultiArrayUtilsTests.swift */ = {isa = PBXFileReference; lastKnownFileType = sourcecode.swift; path = MultiArrayUtilsTests.swift; sourceTree = "<group>"; };
    86  +               7934AF8722EA4CEC00396AD5 /* gpt2-512.mlmodel */ = {isa = PBXFileReference; lastKnownFileType = file.mlmodel; path = "gpt2-512.mlmodel"; sourceTree = "<group>"; };
87  87                  7944CDBE22C501D6000C0B1E /* dev-v1.1.json */ = {isa = PBXFileReference; fileEncoding = 4; lastKnownFileType = text.json; path = "dev-v1.1.json"; sourceTree = "<group>"; };
88  88                  796DF50F22E0EB1D00140C02 /* GPT2Tokenizer.swift */ = {isa = PBXFileReference; lastKnownFileType = sourcecode.swift; path = GPT2Tokenizer.swift; sourceTree = "<group>"; };
89  89                  796DF51322E0EE7800140C02 /* gpt2-merges.txt */ = {isa = PBXFileReference; fileEncoding = 4; lastKnownFileType = text; path = "gpt2-merges.txt"; sourceTree = "<group>"; };
```

```
@@ -126,7 +126,6 @@
```

```
126 126
```

```
                79F2CC9D22C57825009F8551 /* BertForQATests.swift */ = {isa = PBXFileReference; lastKnownFileType = sourcecode.swift; path = BertForQATests.swift; sourceTree = "<group>"; };
127 127         79F2CC9F22C666C7009F8551 /* question_tokens.json */ = {isa = PBXFileReference; fileEncoding = 4; lastKnownFileType = text.json; path = question_tokens.json; sourceTree = "<group>"; };
128 128         79F2CCA122C6717D009F8551 /* LoaderView.swift */ = {isa = PBXFileReference; fileEncoding = 4; lastKnownFileType = sourcecode.swift; path = LoaderView.swift; sourceTree = "<group>"; };
129     -       79F7060F22EA472D00C4432C /* gpt2-256.mlmodel */ = {isa = PBXFileReference; lastKnownFileType = file.mlmodel; path = "gpt2-256.mlmodel"; sourceTree = "<group>"; };
130 129     /* End PBXFileReference section */
131 130
132 131     /* Begin PBXFrameworksBuildPhase section */
        @@ -173,7 +172,7 @@
173 172                                 796DF51322E0EE7800140C02 /* gpt2-merges.txt */,
174 173                                 796DF51422E0EE7800140C02 /* gpt2-vocab.json */,
175 174                                 796DF57D22E1209B00140C02 /* encoded_tokens.json */,
176     -                               79F7060F22EA472D00C4432C /* gpt2-256.mlmodel */,
    175 +                               7934AF8722EA4CEC00396AD5 /* gpt2-512.mlmodel */,
177 176                             );
178 177                             path = Resources;
179 178                             sourceTree = "<group>";
        @@ -445,8 +444,8 @@
445 444                             files = (
446 445                                 796DF55422E1026700140C02 /* ViewController.swift in Sources */,
447 446                                 796DF58722E2727000140C02 /* GPT2.swift in Sources */,
    447 +                               7934AF8822EA4CEC00396AD5 /* gpt2-512.mlmodel in Sources */,
448 448                                 796DF56F22E1039C00140C02 /* SquadDataset.swift in Sources */,
```

```
449     -                                       79F7061022EA472D00C4432C /* gpt2-256.mlmodel in Sources */,
450 449                                         796DF55022E1026700140C02 /* AppDelegate.swift in Sources */,
451 450                                         796DF57222E1039C00140C02 /* Utils.swift in Sources */,
452 451                                         796DF57422E1039C00140C02 /* MLMultiArray+Utils.swift in Sources */,
@@ -464,7 +463,6 @@
464 463                             files = (
465 464                                         796DF56722E1026800140C02 /* CoreMLGPT2Tests.swift in Sources */,
466 465                                         791D169322EA4918004D7A79 /* MultiArrayUtilsTests.swift in Sources */,
467     -                                       79F7061122EA472D00C4432C /* gpt2-256.mlmodel in Sources */,
468 466                             );
469 467                             runOnlyForDeploymentPostprocessing = 0;
470 468                     };
```

CoreMLGPT2Tests/MultiArrayUtilsTests.swift +32 -1

```
@@ -17,8 +17,39 @@ class MultiArrayUtilsTests: XCTestCase {
17 17
18 18           let o = MLMultiArray.slice(arr, sliceDim: 1, selectDims: [0: 0, 2: 0])
19 19           print(o.debug)
20     -
    20 +           XCTAssertEqual(
    21 +               MLMultiArray.toDoubleArray(o),
    22 +               [2, 6, 10]
    23 +           )
21 24           let oo = MLMultiArray.slice(arr, indexing: [.select(0), .slice, .select(0)])
22 25           print(oo.debug)
    26 +           XCTAssertEqual(
    27 +               MLMultiArray.toDoubleArray(oo),
    28 +               [2, 6, 10]
    29 +           )
```

```
+     }
+
+     func testTopK() {
+         let arr: [Double] = Array(0..<4).map { Double($0) } + Array(30..<36).map { Double($0) }
+         XCTAssertEqual(
+             arr,
+             [0.0, 1.0, 2.0, 3.0, 30.0, 31.0, 32.0, 33.0, 34.0, 35.0]
+         )
+         let topk = Math.topK(arr: arr, k: 6)
+         XCTAssertEqual(
+             topk.indexes,
+             [9, 8, 7, 6, 5, 4]
+         )
+         XCTAssertEqual(
+             topk.probs,
+             [0.6336913, 0.233122, 0.085760795, 0.031549633, 0.011606461, 0.0042697783]
+         )
+
+         let sampleIndex = Math.sample(indexes: topk.indexes, probs: topk.probs)
+         print("sampleIndex", sampleIndex)
+
+         XCTAssertTrue(
+             sampleIndex >= 4 && sampleIndex <= 9
+         )
      }
  }
```

Sources/Math.swift +78

```
@@ -54,4 +54,82 @@ struct Math {
             let ptr = UnsafeMutablePointer<Double>(OpaquePointer(multiArray.dataPointer))
             return Math.argmax(ptr, count: multiArray.count)
         }
+
+     /// Helper for top-k
```

```
59 +     ///
60 +     static func topK(arr: [Double], k: Int) -> (indexes: [Int], probs:
   [Float]) {
61 +         let x = Array(arr.enumerated().map { ($0, $1) }
62 +             .sorted(by: { a, b -> Bool in a.1 > b.1 })
63 +             .prefix(through: min(k, arr.count) - 1))
64 +         let indexes = x.map { $0.0 }
65 +         let logits = x.map { Float($0.1) }
66 +         let probs = softmax(logits)
67 +         return (indexes: indexes, probs: probs)
68 +     }
69 +
70 +     /// Multinomial sampling from an array of probs. Works well with topK
71 +     static func sample(indexes: [Int], probs: [Float]) -> Int {
72 +         let i = randomNumber(probabilities: probs)
73 +         return indexes[i]
74 +     }
75 +
76 +     /**
77 +      Computes the "softmax" function over an array.
78 +      Based on code from https://github.com/nikolaypavlov/MLPNeuralNet/
79 +      This is what softmax looks like in "pseudocode" (actually using
   Python
80 +      and numpy):
81 +      x -= np.max(x)
82 +      exp_scores = np.exp(x)
83 +      softmax = exp_scores / np.sum(exp_scores)
84 +      First we shift the values of x so that the highest value in the
   array is 0.
85 +      This ensures numerical stability with the exponents, so they don't
   blow up.
86 +      */
87 +     static func softmax(_ x: [Float]) -> [Float] {
88 +         var x = x
89 +         let len = vDSP_Length(x.count)
90 +
91 +         // Find the maximum value in the input array.
92 +         var max: Float = 0
93 +         vDSP_maxv(x, 1, &max, len)
```

```
94  +
95  +         // Subtract the maximum from all the elements in the array.
96  +         // Now the highest value in the array is 0.
97  +         max = -max
98  +         vDSP_vsadd(x, 1, &max, &x, 1, len)
99  +
100 +         // Exponentiate all the elements in the array.
101 +         var count = Int32(x.count)
102 +         vvexpf(&x, x, &count)
103 +
104 +         // Compute the sum of all exponentiated values.
105 +         var sum: Float = 0
106 +         vDSP_sve(x, 1, &sum, len)
107 +
108 +         // Divide each element by the sum. This normalizes the array
    contents
109 +         // so that they all add up to 1.
110 +         vDSP_vsdiv(x, 1, &sum, &x, 1, len)
111 +
112 +         return x
113 +     }
114 +
115 +     /// Multinomial sampling
116 +     ///
117 +     /// From https://stackoverflow.com/questions/30309556/generate-
    random-numbers-with-a-given-distribution
118 +     ///
119 +     static func randomNumber(probabilities: [Float]) -> Int {
120 +         // Sum of all probabilities (so that we don't have to require
    that the sum is 1.0):
121 +         let sum = probabilities.reduce(0, +)
122 +         // Random number in the range 0.0 <= rnd < sum :
123 +         let rnd = sum * Float(arc4random_uniform(UInt32.max)) /
    Float(UInt32.max)
124 +         // Find the first interval of accumulated probabilities into
    which `rnd` falls:
125 +         var accum: Float = 0.0
126 +         for (i, p) in probabilities.enumerated() {
127 +             accum += p
```

```
+                if rnd < accum {
+                    return i
+                }
+            }
+            // This point might be reached due to floating point inaccuracies:
+            return (probabilities.count - 1)
+        }
    }
```

Comments 0 (0)

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